# tides

## _travelers guide
## _for managing gravity

| 3 modes | 3 ranges | 20 min to 10 khz |
|---|---|---|

**[ range ]** press  **[ range ]** hold  › then  noon = 12 o'clock  cw / ccw = clockwise / counter

one envelope, lfo or oscillator · shape, slope and smoothness bend the wave in three directions

the 2014 tides, firmware 1.1 · long press = 1 s

magenta = hold

## 01 buttons · LEDs

td_01

| | |
|---|---|
| **[ mode ]** | AD envelope ● › cycle ○ › AR envelope ● › cycle … steps back and forth |
| **[ range ]** | low ● 20 min–5 hz › medium ○ 0.05–300 hz › high ● 8 hz–10 khz, back and forth |
| **[ range ]** 1 s | **clocked / pll mode** on or off · the range LED blinks while it's on (06) |
| **[ mode ]** 1 s | **calibration** · only with cables in the fm and level inputs (08) |
| output LED | brightness = the wave · ● attack · ● decay, sustain, release |
| remembered | mode, range and clocked mode survive power off |

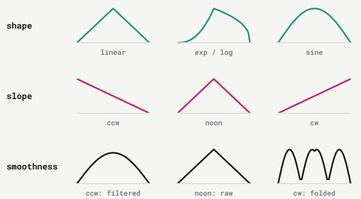
## 02 knobs

td_02

| | |
|---|---|
| **[ frequency ]** | ±4 octaves around the range · in clocked mode it picks the ratio instead |
| **[ fm ]** att | polarity and amount of the fm input · unpatched = fine tune |
| **[ shape ]** | the curve of the rise and the fall: linear, exponential or sine segments |
| **[ slope ]** | rise vs fall time · ccw = no rise · noon = equal · cw = no fall |
| **[ smoothness ]** | noon = raw · ccw = 2-pole low-pass, rounder · cw = wavefolder, bumps and kinks |
| cv ins | shape, slope and smoothness jacks add to their knobs |

## __fig.01 : three dimensions

the wave, bent




three knobs, three directions · each has its own cv input

## 03 modes

td_03 · **[ mode ]**

| | |
|---|---|
| ● AD | **single shot**: rise then fall on each **[ trig ]** · needs a trigger to move |
| ○ cycle | **looping**: lfo at low / medium, oscillator at high · **[ trig ]** resets the phase |
| ● AR | **sustained**: rises, holds while the **[ trig ]** gate is high, falls when it ends |
| high range | a different set of shapes, a linear (pwm-like) slope, built to sound different at audio rate |

## 04 inputs

td_04 · all 100k

| | |
|---|---|
| **[ trig ]** | AD / AR: the trigger or gate · cycle: phase reset · 0.7 v threshold |
| **[ freeze ]** | high = the wave stops where it is, until the input drops |
| **[ v/oct ]** | pitch, digitally tracked · calibrate if it drifts |
| **[ fm ]** | through the fm attenuverter · read at 6 khz |
| **[ level ]** | a vca on uni and bi, 0–8 v · unpatched = full level |
| **[ clock ]** | locks the frequency or envelope time to a clock or audio (06) |

## 05 outputs

td_05

| | |
|---|---|
| **[ uni ]** | 0 to 8 v: the wave as you'd expect it |
| **[ bi ]** | ±5 v, but a different wave: a bump up then a bump down (fig.02) · slope sets the ratio between them |
| **[ high tide ]** | +5 v from the end of the rise until the cycle restarts or retriggers |
| **[ low tide ]** | +5 v from the end of the fall until the next trigger · chain envelopes with either |
| try | crossfade uni and bi at audio rate: a fourth dimension of timbre |

## 09 peacocks

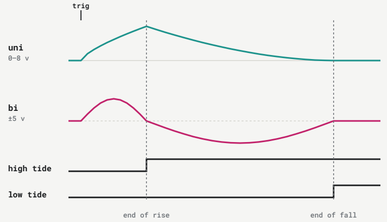
td_09 · the easter egg

1. dummy cables in **[ fm ]** and **[ level ]**
2. hold **[ mode ]** 1 s › yellow · hold **[ mode ]** 1 s again › green
3. hold **[ range ]** 1 s › red

uni and bi into a scope in x–y mode draw a pair of peacocks · any button leaves

## __fig.02 : uni, bi and the tides

one AD envelope




## 06 clocked · pll

td_06 · **[ range ]** 1 s

| | |
|---|---|
| **[ range ]** 1 s | on: tides locks to the **[ clock ]** input · the range LED blinks · same again = off |
| **[ frequency ]** | now picks the ratio to the clock: divide ccw, multiply cw, 1:1 at noon (fig.04) |
| audio in | feed it an audio signal and it plays a harmony locked to it |

## __fig.04 : clock ratios

the frequency knob, clocked




frequency knob in clocked mode · noon = 1:1

## 10 the high range

td_10 · time vs frequency

| | |
|---|---|
| why | at audio rate you hear harmonics, not the drawn wave, so high mode swaps in shapes that sound different |
| **[ slope ]** | linear here, like pwm · exponential-ish in low and medium, for very short attacks and long decays |
| expect | small level and offset errors: some edges are smoothed to stop aliasing |

## 12 the module

td_12 · 2014–2018

| | |
|---|---|
| size | 14 hp · 25 mm deep · 55 mA +12 v · 5 mA –12 v |
| range | 0.5 mhz to 10 khz across the three ranges |
| cv | shape, slope, smooth read at 1 khz · fm 6 khz · level 48 khz · outs 16-bit, 48 khz |

## 07 calibration

td_07

### v/oct, fm and level

if notes drift out of tune

1. **[ frequency ]** at noon · a cable in **[ fm ]** and one in **[ level ]**, other ends loose
2. keyboard or precision cv into **[ v/oct ]**
3. hold **[ mode ]** 1 s › all LEDs yellow
4. send 1 v (c2) · press **[ mode ]** › all LEDs green
5. send 3 v (c4) · press **[ mode ]** › done

### bi output offset

trimmer on the back

1. nothing in **[ trig ]** · mode AD or AR · range medium
2. measure **[ bi ]** and turn the trimmer to 0 v

## 08 firmware

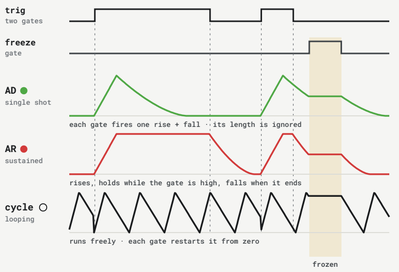
td_08 · v1.1, the last

1. unplug everything · audio out of your computer into **[ fm ]**
2. power on holding **[ mode ]** · all 3 LEDs blink yellow
3. play tides_1.1.wav · LEDs cycle and flash green per block · it reboots when done

all LEDs red = level too low or high: change it, press **[ mode ]**, start the file again · 1.1 added an alias-free (polyblep) triangle at audio rate

## __fig.05 : one gate, three modes

trig + freeze




## 11 first patches

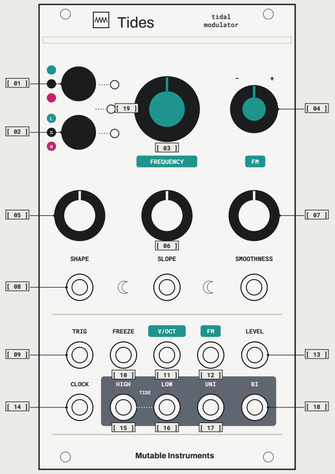
td_11

**percussive envelope**
● AD · medium · trigger › **[ trig ]** · **[ uni ]** › a vca
**[ slope ]** ccw = instant attack, long decay

**voice**
○ cycle · high · pitch › **[ v/oct ]**
**[ smoothness ]** is the filter / folder · crossfade uni and bi for timbre

## __fig.03 : front panel

01–08 = controls · 09–19 = jacks + LEDs




| | |
|---|---|
| [ 01 ] | **__mode :** AD · cycle · AR |
| [ 02 ] | **__range :** low · med · high · clock |
| [ 03 ] | **__frequency :** ±4 oct · clock ratio |
| [ 04 ] | **__fm att :** fine tune unpatched |
| [ 05 ] | **__shape :** segment curves |
| [ 06 ] | **__slope :** rise vs fall |
| [ 07 ] | **__smoothness :** filter ‹ › fold |
| [ 08 ] | **__cv ins :** shape · slope · smooth |

| | |
|---|---|
| [ 09 ] | **__trig :** trigger · gate · reset |
| [ 10 ] | **__freeze :** stops the wave |
| [ 11 ] | **__v/oct :** pitch |
| [ 12 ] | **__fm :** through 04 · firmware in |
| [ 13 ] | **__level :** vca for uni / bi |
| [ 14 ] | **__clock :** sync · pll |
| [ 15 ] | **__high tide :** end of rise |
| [ 16 ] | **__low tide :** end of fall |
| [ 17 ] | **__uni :** 0–8 v |
| [ 18 ] | **__bi :** ±5 v, two bumps |
| [ 19 ] | **__LEDs :** mode · output · range |

## _where things hide

no menus · td_13

### held buttons

- **range 1 s** clocked / pll on · off
- **mode 1 s** calibration (cables in fm + level)
- **mode 1 s ×2, range 1 s** the peacocks, from calibration

### at power on

- **hold mode** firmware update, audio into fm
- **remembered** mode, range, clocked mode

### LED codes

top to bottom: mode, output, range

| | |
|---|---|
| **mode** | ● AD ○ cycle ● AR |
| **range** | ● low ○ medium ● high **blink** clocked |
| **output** | ● rising ● falling / held brightness = level |
| **calibration** | ● send 1 v ● send 3 v |
| **firmware** | ● blinking = ready ● flash = block ok ● all = level wrong |

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